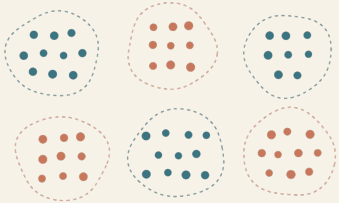
Adaptive Design of Clustered Surveys and Experiments

Carlos Gonzalez-Perez, University of Oxford



Outline

Introduction and Motivation

Model and Study Policies

Adaptive Policies

Conditional Universality

Application

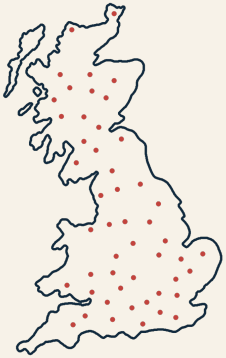
Conclusion

A Survey in Great Britain



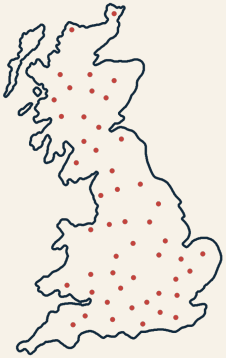
Sedgwick, P. (2015). Multistage sampling

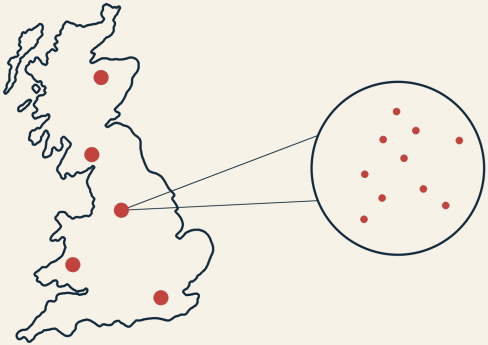
Clustered Designs



Sedgwick, P. (2015). Multistage sampling

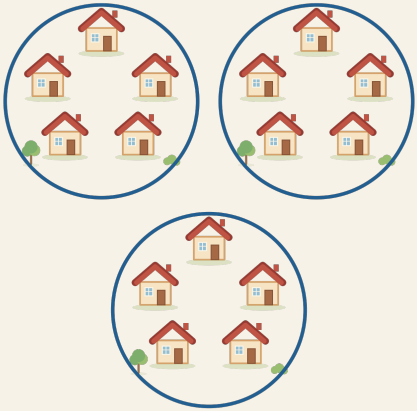
Clustered Designs





Sedgwick, P. (2015). Multistage sampling

More Clusters or More Units per Cluster?



Optimal Allocation

- **trade-off** between **sampling costs** and **dependence of units within clusters**
- for a given **budget**, the **optimal** (variance minimizing) allocation depends on
  - cost structure, usually **known**
  - **intra-cluster correlation** (ICC), usually **unknown**
- **Guessing is a bad idea:** Knox and Chondros (2004) report variations of the ICC of one order of magnitude within the same population

Research Questions

- **Can we design optimal clustered surveys and experiments under unknown ICC?**

- **Can we do so without compromising inference?**

Performance

- **Can we design optimal clustered surveys and experiments under unknown ICC?**
- Two **pilot-based algorithms**
  - sample clusters **sequentially** and **learn** the ICC while the experiment runs
  - implement the **optimal design** in the limit
- **Great properties**
  - **negligible** excess variance
  - **beat any static design** by an order of magnitude
  - **rate unimprovable** (minimax optimal)

Revisiting Fink, Jack and Masiye (2020, AER)

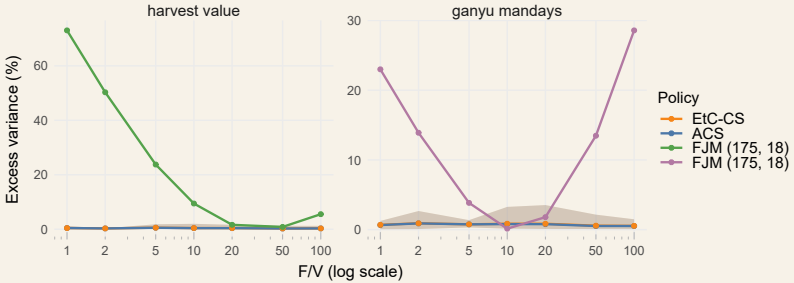
- **Setting:** favorable loans to farmers in Zambia with treatment at the **village** level
  - assess loan impact on harvest output and *ganyu* labor demand
  - 175 villages (clusters) $\times$ 18 households per village (units)

Revisiting Fink, Jack and Masiye (2020, AER)

- **Setting:** favorable loans to farmers in Zambia with treatment at the **village** level
  - assess loan impact on harvest output and *ganyu* labor demand
  - 175 villages (clusters) $\times$ 18 households per village (units)

- Was the realized design **cost-optimal?**
  - I don't observe the cost ratio F/V, but possibly $F/V \in [1, 2]$
  - FJM design only optimal at $F/V \approx 33$ for harvest (and ≈ 12 for ganyu)
  - away from these ratios, realized design carries **up to 70% excess variance**

What Would Adaptive Allocation Have Delivered?



Notes: Excess variance is presented relative to the oracle design (%). Lines are means over 10,000 simulations, bands are 95% intervals across simulations.

- My algorithms achieve **up to 41%** **lower variance** at equal cost or achieve equal precision at **up to 42%** **lower cost** and are **robust to ICC misspecification**

Conditional Inference

- **Design depends on sample data**
  - Distribution of terminal estimator **conditional on the design**?

Conditional Inference

- **Design depends on sample data**
  - Distribution of terminal estimator **conditional on the design**?
- **Can we do so without compromising inference?**

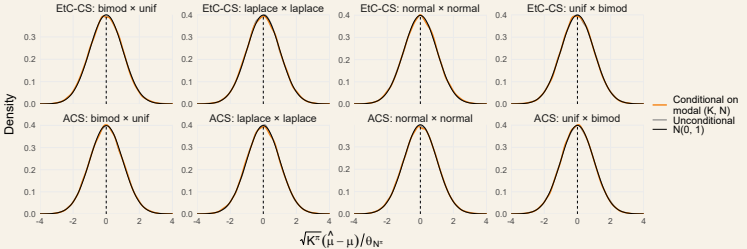
Conditional Inference

- **Design depends on sample data**
  - Distribution of terminal estimator **conditional on the design**?
- **Can we do so without compromising inference?**
- **YES** under **error symmetry**
  - designs depending on variances leave mean estimators **conditionally normal**
  - **without correction nor data exclusion**

Conditional Inference

- **Design depends on sample data**
  - Distribution of terminal estimator **conditional on the design**?
- **Can we do so without compromising inference?**
- **YES** under **error symmetry**
  - designs depending on variances leave mean estimators **conditionally normal**
  - **without correction nor data exclusion**
- **General principle**
  - Neyman allocations, PropScores-based treatment allocations (Hahn et al., 2011)

Conditional Inference is Asymptotically Normal



Notes: Empirical density at the modal design. Distribution of u_k (left), of ε_{ki} (right). Distributions considered: Bimodal-Uniform, Laplace-Laplace, Normal-Normal, Uniform-Bimodal. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$.

- **We were doing fine and we did not know!** (under symmetry and policy evenness)

Outline

Introduction and Motivation

Model and Static Policies

Adaptive Policies

Conditional Inference

Application

Conclusion

Model – Survey Design Problem

- Extends mechanically to experimental design and ATE estimation

- Nested **random effects** model [***A1***]

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \qquad \mathrm{Var}(u_k) = \tau^2, \quad \mathrm{Var}(\varepsilon_{ki}) = \sigma^2, \qquad u_k \perp\!\!\!\perp \varepsilon_{ki}, \quad iid$$

  - **unknown variance ratio** $\lambda_0 = \sigma^2/\tau^2$, $\mathrm{ICC} = 1/(1+\lambda_0)$
  - $\omega = (\tau^2, \lambda_0) \in \Omega$ with Ω **known** and **compact** [***A2***], with $\lambda_0 \in [\lambda_m, \lambda_M]$

Model – Survey Design Problem

- Define the **average of averages** estimator

$$\hat{\mu}_{K,N} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{N}\sum_{i=1}^{N} y_{ki}, \qquad K \text{ clusters, } N \text{ number of units}$$

- **Goal**

$$\min_{K,N} \ \mathrm{Var}(\hat{\mu}_{K,N}) = W(K,N) \qquad \text{st} \quad K(F+VN) \leq B \ \text{ a.s.}$$

  - **Linear Costs [*A3*]:** F fixed cost/cluster, V variable cost/unit, B fixed budget
  - $W(K,N) = \frac{\tau^2}{K}\left(1+\frac{\lambda_0}{N}\right)$

Cochran's Allocation and Guessing

- **Oracle** solution goes back to Cochran (1977)

$$N^*(\lambda_0) = \sqrt{\lambda_0 F/V}, \qquad K^*(\lambda_0) = \frac{B}{F + VN^*(\lambda_0)}$$

  - **Optimal map** $\lambda \mapsto (K^*(\lambda), N^*(\lambda))$
  - Assume **interior solution** of $(K^*(\lambda), N^*(\lambda))$ for $\lambda \in [\lambda_m, \lambda_M]$ [***A4***]

Cochran's Allocation and Guessing

- **Oracle** solution goes back to Cochran (1977)

$$N^*(\lambda_0) = \sqrt{\lambda_0\, F/V}, \qquad K^*(\lambda_0) = \frac{B}{F + V N^*(\lambda_0)}$$

  - **Optimal map** $\lambda \mapsto (K^*(\lambda), N^*(\lambda))$
  - Assume **interior solution** of $(K^*(\lambda), N^*(\lambda))$ for $\lambda \in [\lambda_m, \lambda_M]$ [***A4***]

- λ_0 is **unknown**, so in practice, designers **guess**
  - guessing is **difficult** Knox and Chondros (2004)
  - guessing is **expensive**, the excess variance coming from ICC misimputation is of the same order as the target variance

Literature Review

- **Survey Theory** *mostly* ignores ICC uncertainty
  - prominent role of ICC, but rarely estimated **inside** the optimization problem
  - algorithmic extensions of Cochran's principles (Bethel, 1989; Ballin and Barcaroli, 2013; Barcaroli et al., 2022)

Literature Review

- **Survey Theory** *mostly* ignores ICC uncertainty
  - prominent role of ICC, but rarely estimated **inside** the optimization problem
  - algorithmic extensions of Cochran's principles (Bethel, 1989; Ballin and Barcaroli, 2013; Barcaroli et al., 2022)
- **Statistics and Biostatistics** deals with ICC uncertainty
  - **Minimax** over ICC uncertainty (van Breukelen and Candel, 2015, 2018, 2021)
    - adaptive designs dominate static policies by an order of magnitude
  - **Sample-size re-estimation** with nuisance parameters (Stein, 1945)
    - ICC re-estimation (Lake et al., 2002; van Schie and Moerbeek, 2014) no cost structure, no optimality theory, a single margin (width or depth)

Literature Review

- **Survey Theory** *mostly* ignores ICC uncertainty
  - prominent role of ICC, but rarely estimated **inside** the optimization problem
  - algorithmic extensions of Cochran's principles (Bethel, 1989; Ballin and Barcaroli, 2013; Barcaroli et al., 2022)
- **Statistics and Biostatistics** deals with ICC uncertainty
  - **Minimax** over ICC uncertainty (van Breukelen and Candel, 2015, 2018, 2021)
    - adaptive designs dominate static policies by an order of magnitude
  - **Sample-size re-estimation** with nuisance parameters (Stein, 1945)
    - ICC re-estimation (Lake et al., 2002; van Schie and Moerbeek, 2014) no cost structure, no optimality theory, a single margin (width or depth)
- First paper to establish **optimality guarantees** against the oracle under **unknown ICC** in the **joint allocation** problem

Excess Variance and the Cost of Guessing

- **Oracle variance**

$$W_0 = W(K^*(\lambda_0), N^*(\lambda_0)) = \boldsymbol{\Theta(1/B)}$$

Excess Variance and the Cost of Guessing

- **Oracle variance**

$$W_0 = W(K^*(\lambda_0), N^*(\lambda_0)) = \boldsymbol{\Theta(1/B)}$$

- **Excess variance** (neyman regret)

$$\Delta_\pi = W(K^\pi, N^\pi) - W_0 \geq 0, \qquad \max_{\omega \in \Omega} \mathbb{E}_{\omega,\,\pi}[\Delta_\pi]$$

Excess Variance and the Cost of Guessing

- **Oracle variance**

$$W_0 = W(K^*(\lambda_0), N^*(\lambda_0)) = \boldsymbol{\Theta(1/B)}$$

- **Excess variance** (neyman regret)

$$\Delta_\pi = W(K^\pi, N^\pi) - W_0 \geq 0, \qquad \max_{\omega \in \Omega} \mathbb{E}_{\omega,\,\pi}[\Delta_\pi]$$

- **Cost of guessing** in static policies (Lemma 5.2)

$$\Delta_{\text{guessing}} = W(K^*(\lambda), N^*(\lambda)) - W_0 = \boldsymbol{\Theta(1/B)} \text{ for all } \lambda \neq \lambda_0$$

  - **guessing is expensive**, same order as the oracle variance itself
  - true for **any** static λ, including minimax λ (van Breukelen and Candel, 2015)

Outline

Introduction and Motivation

Global and Static Policies

Adaptive Policies

Conditional Updates

Application

Conclusion

Pilot-Based Algorithms and Implementability

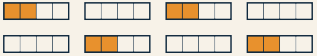


population

Pilot-Based Algorithms and Implementability



population



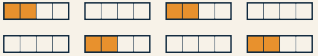
pilot $(K_p, N_p) = (4, 2)$

$\hat{\lambda}_p = \hat{\lambda}_{K_p, N_p}$

Pilot-Based Algorithms and Implementability



population



pilot $(K_p, N_p) = (4, 2)$

$\hat{\lambda}_p = \hat{\lambda}_{K_p, N_p}$



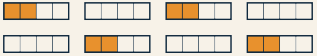
final design $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$

$= (5, 3)$

Pilot-Based Algorithms and Implementability



population



pilot $(K_p, N_p) = (4, 2)$

$\hat{\lambda}_p = \hat{\lambda}_{K_p, N_p}$



final design $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$

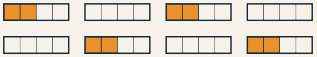
$= (5, 3)$

- **How to select the pilot?** More clusters $\implies$ better $\hat{\lambda}_p$, but ...

Pilot-Based Algorithms and Implementability



population



pilot $(K_p, N_p) = (4, 2)$

$\hat{\lambda}_p = \hat{\lambda}_{K_p, N_p}$



final design $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$

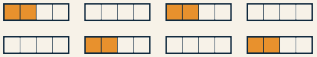
$= (5, 3)$

- **How to select the pilot?** More clusters $\implies$ better $\hat{\lambda}_p$, but ...
- What if $K_p > K^*(\hat{\lambda}_p)$? i.e. final design had been 3×4? **Cluster oversampling**

Pilot-Based Algorithms and Implementability

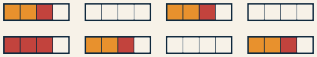


population



pilot $(K_p, N_p) = (4, 2)$

$\hat{\lambda}_p = \hat{\lambda}_{K_p, N_p}$



final design $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$

$= (5, 3)$

- **How to select the pilot?** More clusters $\implies$ better $\hat{\lambda}_p$, but ...
- What if $K_p > K^*(\hat{\lambda}_p)$? i.e. final design had been 3×4? **Cluster oversampling**
- **Circularity!** Optimal design requires ICC, but learning ICC requires many clusters

Escaping the Circularity through Adaptivity

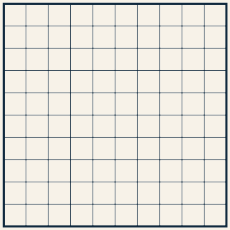
- **Explore-then-Commit Cluster Selection** (`EtC-CS`)
  - **Smallest plausible pilot**

$$(K_p, N_p) = (K^*(\lambda_M), N^*(\lambda_m)) \implies$$
$$K^*(\lambda_M) \le K^*(\hat{\lambda}_p) \text{ and } N^*(\lambda_m) \le N^*(\hat{\lambda}_p) \quad \forall\ \hat{\lambda}_p \in [\lambda_m, \lambda_M]$$

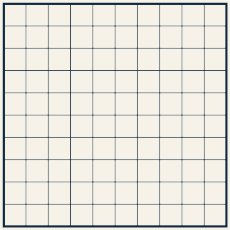
  - Implementable by design/assumption **[*A2*]**, possibly overly conservative

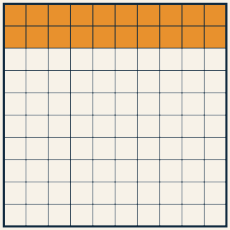
- `EtC-CS` is already **minimax optimal** (Theorem 5.4)

Why Committing so Early?

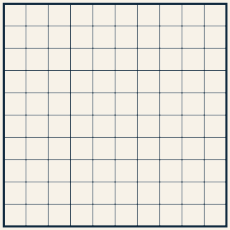


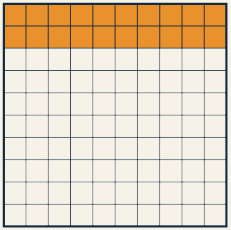
Why Committing so Early?

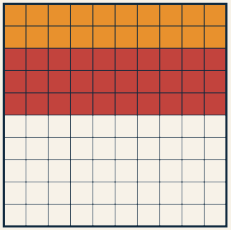




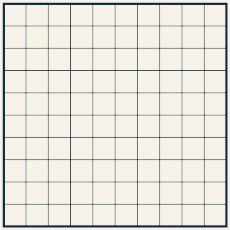
Why Committing so Early?

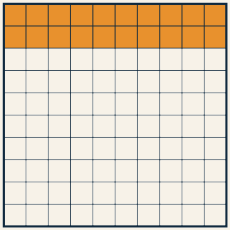


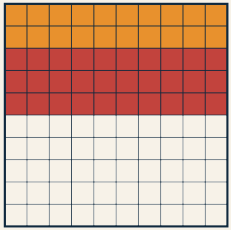


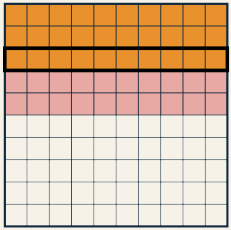


Why Committing so Early?

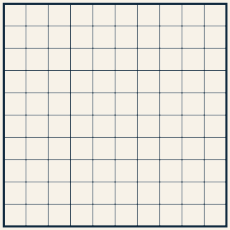


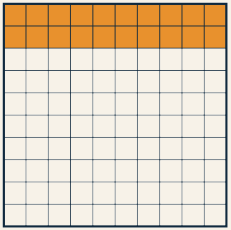


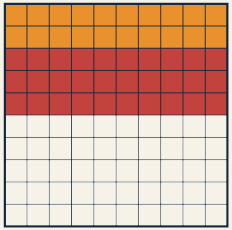


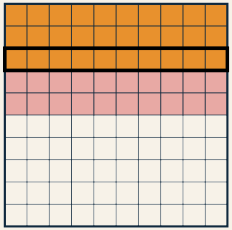


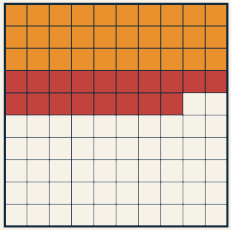
Why Committing so Early?









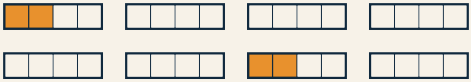


Adaptive Cluster Selection (ACS)

- fix shallow $N_p = N^*(\lambda_m)$ and **sample one cluster at a time**
- **stop before oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$

Adaptive Cluster Selection (ACS)

- fix shallow $N_p = N^*(\lambda_m)$ and **sample one cluster at a time**
- **stop before oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$

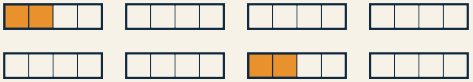


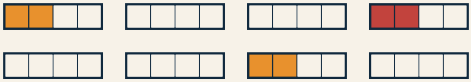
$$(K^*(\hat{\lambda}_{2,2}), N^*(\hat{\lambda}_{2,2})) = (5, 2)$$

$$2 + 1 = 3 \leq 5 \checkmark$$

Adaptive Cluster Selection (ACS)

- fix shallow $N_p = N^*(\lambda_m)$ and **sample one cluster at a time**
- **stop before oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$



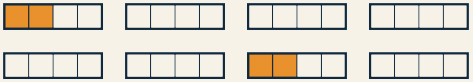


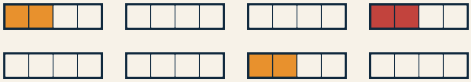
$$(K^*(\hat{\lambda}_{3,2}), N^*(\hat{\lambda}_{3,2})) = (4, 3)$$

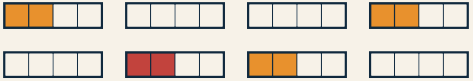
$$3 + 1 = 4 \leq 4 \checkmark$$

Adaptive Cluster Selection (ACS)

- fix shallow $N_p = N^*(\lambda_m)$ and **sample one cluster at a time**
- **stop before oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$





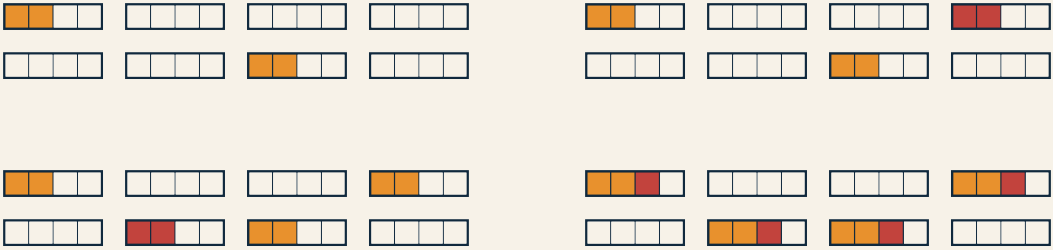


$$(K^*(\hat{\lambda}_{4,2}), N^*(\hat{\lambda}_{4,2})) = (4, 3)$$

$$4 + 1 = 5 \leq 4 \text{ ✗}$$

Adaptive Cluster Selection (ACS)

- fix shallow $N_p = N^*(\lambda_m)$ and **sample one cluster at a time**
- **stop before oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$



complete final design

$(K^{\texttt{ACS}}, N^{\texttt{ACS}}) = (4, 3)$

Theorems 5.4 and 5.5. Upper and Lower Bounds

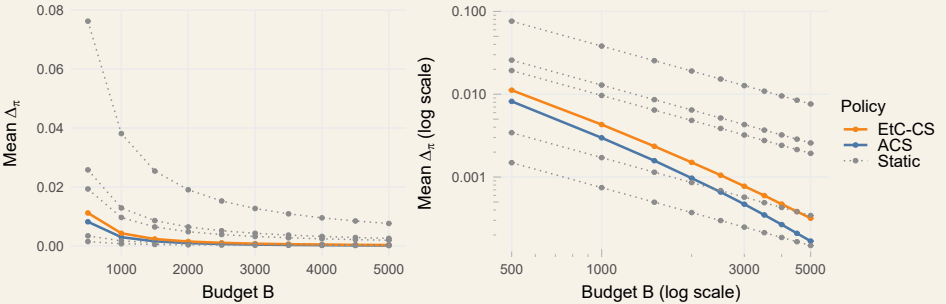
Under $[A1 - A4]$ + subgaussian u_k, ε_{ki} $[A5]$, $\exists$ constants $C_\pi < \infty$ not dep. on B

$$\max_{\omega \in \Omega} \mathbb{E}\left[\Delta_{\texttt{Etc-cs}}\right] \leq \frac{C_{\texttt{Etc-cs}}}{B^2}, \qquad \max_{\omega \in \Omega} \mathbb{E}\left[\Delta_{\texttt{ACS}}\right] \leq \frac{C_{\texttt{ACS}}}{B^2}$$

Moreover, for every policy π there exists $\omega \in \Omega$ such that $\mathbb{E}[\Delta_\pi] \geq C/B^2$.

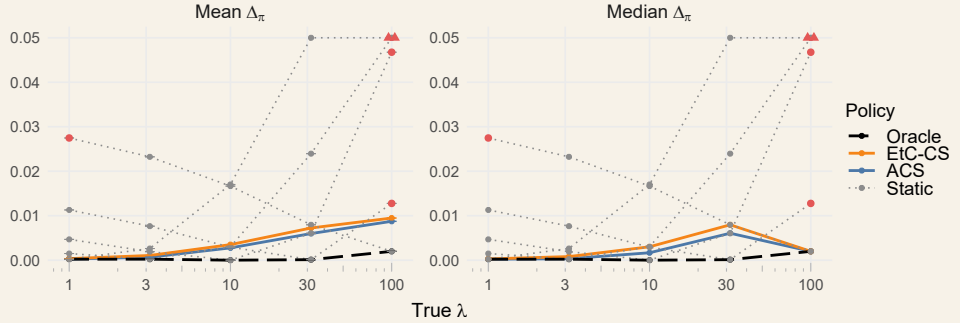
- asymptotically **negligible** excess variance, $O(1/B^2) << \Theta(1/B)$
- algorithms asymptotically dominate every **misspecified static policy**
- matching bounds hence **minimax rate optimal**

Simulations – Excess Variance across Budget



Notes: Static policies at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $F = 20$, $V = 2$.

Simulations – Excess Variance across λ_0



Notes: Static policies at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $F = 20$, $V = 2$.

Outline

Introduction and Motivation

Global and Static Policies

Adaptive Policies

Conditional Inference

Application

Conclusion

Does Adaptivity Compromise Inference? Two Traditions

- (K^π, N^π) depends on **pilot data** $\Rightarrow$ is $\hat{\mu} \mid (K^\pi, N^\pi)$ centered or asympt normal?
  - adaptive sampling biases estimates and destroys normality
  - (Nie et al., 2018; Chen and Andrews, 2023; Wager, 2024)

Does Adaptivity Compromise Inference? Two Traditions

- (K^π, N^π) depends on **pilot data** $\Rightarrow$ is $\hat{\mu} \mid (K^\pi, N^\pi)$ centered or asympt normal?
  - adaptive sampling biases estimates and destroys normality
  - (Nie et al., 2018; Chen and Andrews, 2023; Wager, 2024)
- **Adjustment tradition:** restore normality by **reweighting** the observations
  - Hadad et al. (2021); Zhang et al. (2021); Yin and Shen (2005)

Does Adaptivity Compromise Inference? Two Traditions

- (K^π, N^π) depends on **pilot data** $\Rightarrow$ is $\hat{\mu} \mid (K^\pi, N^\pi)$ centered or asympt normal?
  - adaptive sampling biases estimates and destroys normality
  - (Nie et al., 2018; Chen and Andrews, 2023; Wager, 2024)
- **Adjustment tradition:** restore normality by **reweighting** the observations
  - Hadad et al. (2021); Zhang et al. (2021); Yin and Shen (2005)
- **Invariance tradition:** under **exact normality** designs driven by **sample variances** do not disturb inference on means
  - Stein (1945); Rinott (1978); Dudewicz and Dalal (1975)
  - in fact, designs driven by **residuals**

Inference Invariance under Symmetry

- This paper **invariance tradition**, relaxing **normality** by **error symmetry** (much lighter nonparametric assumption!)

- **Small cost:** from policies reading residuals, to policies reading **squared errors (even)** or **squared residuals (asymptotically even)**
  - sample variances are functions of squared errors
  - most reasonable policies are even anyway

- Even under gaussianity, **no policy can read levels**

A General Principle

- Mean estimators remain asymptotically **normal** after conditioning on a design
  - generated by any **asymptotically even** function of the data
  - **arbitrary** sequential designs
  - ICC is a function of variances, hence `EtC-CS` and `ACS` are covered
- Theory **extends** **well beyond** my algorithms and the cluster setting
  - **Neyman allocations** and **optimal propensity scores** (Hahn et al., 2011)
  - allows conditioning on **non-degenerate events** (Hahn et al., 2011)
  - avoids **pilot data exclusion** (Viviano, 2022)

Why Symmetry? Signs and Magnitudes

- Under symmetry, cluster errors **factor** into

$$\text{signs} \times \text{magnitudes} \qquad \text{with signs} \perp\!\!\!\perp \text{magnitudes}$$

- Even functions of the data are functions of **magnitudes alone**
  - conditioning on the magnitudes **freezes** the design ...
  - ... but leaves signs **untouched**, so a **CLT can run through them**

Asymptotic Evenness

- In practice, policies read **sample** variances
  - demean by the sample mean $\Rightarrow$ designs depend on the **error signs**
- **Strategy:** Nominate an infeasible **surrogate even** policy
  - designs generated by the asymp even policy $\xrightarrow{p}$ designs of even surrogate
  - events depending on asympt even designs $\xrightarrow{p}$ events depending on even designs
- **Converging Scale?**
  - **unit** $\quad | K^{\tilde{\pi}} - K^{\pi} | \xrightarrow{p} 0$
  - $\boldsymbol{o_p(B)}$ $\quad | K^{\tilde{\pi}}/B - K^{\pi}/B | \xrightarrow{p} 0$
  - $\boldsymbol{o_p(\sqrt{B})}$ $\quad | K^{\tilde{\pi}}/\sqrt{B} - K^{\pi}/\sqrt{B} | \xrightarrow{p} 0$ – **asymptotically even**

Why $O_p(\sqrt{B})$?

- **Feasibility**
  - replacing sq. errors by sq. residuals usually **preserves** $o_p(\sqrt{B})$ coupling, and faster like $o_p(B)$
  - unit coupling is usually **infeasible**

Why $O_p(\sqrt{B})$?

- **Feasibility**
  - replacing sq. errors by sq. residuals usually **preserves** $o_p(\sqrt{B})$ coupling, and faster like $o_p(B)$
  - unit coupling is usually **infeasible**

- **Non-degeneracy of conditioning events**
  - recall $K^*(\lambda_0) = B \cdot h(\lambda_0)$ and so consistent $K^\pi/B \xrightarrow{p} h(\lambda_0)$
  - all events depending on K^π/B are **degenerate** (conditional = unconditional inf)
  - while designs involving $K^\pi/\sqrt{B}$ are **non-degenerate**, like K^π quantiles

Summary Scale Conditioning Events

Scale	**Feasible Surrogate Coupling?**	**Non-degenerate Events?**	**Tools**
Unit	✗	–	–
$1/B$	✓	✗	Anscombe
$1/\sqrt{B}$ (new)	✓	✓	Symmetry and Evenness

$\sqrt{B}$ is the *only* feasible scale with non-degenerate events

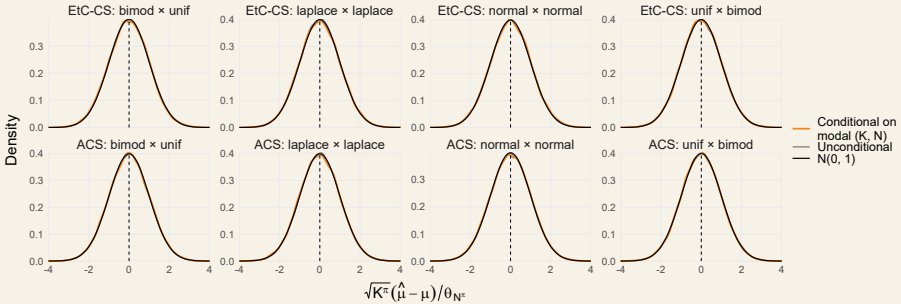
Theorem 6.10. Design-Robust Inference for `EtC-CS` and `ACS`

Under $[A1-A4] + (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots) \overset{d}{=} -(u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ $[A6]$,
for any design event A_B (i) depending on $K^\pi/\sqrt{B}$, (ii) with $\lim_{B\to\infty} \mathbb{P}(A_B) > 0$, and (iii) satisfying an anti-concentration condition,

$$\mathcal{L}\left(\frac{\sqrt{K^\pi}\,(\hat{\mu}_{K^\pi,\,N^\pi} - \mu)}{\hat{b}_{K^\pi,\,N^\pi}} \;\middle|\; A_B \right) \Longrightarrow N(0,1) \qquad \pi \in \{\texttt{EtC-CS}, \texttt{ACS}\}$$

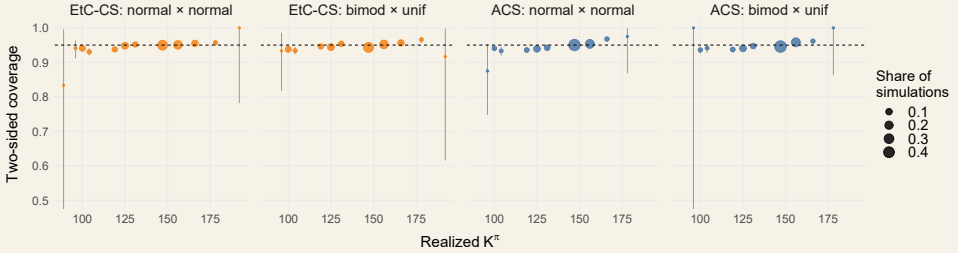
- **Nothing needs to change:** full sample, usual clustered SEs and normal CV—no reweighting, no sample splitting, no discarded data

Simulations – Inference is Unaffected



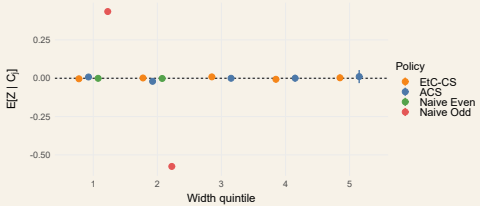
Notes: Empirical density at the modal design. Distribution of u_k (left), of ε_{ki} (right). Distributions considered: Bimodal-Uniform, Laplace-Laplace, Normal-Normal, Uniform-Bimodal. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$.

Simulations – Cell-Conditional Inference

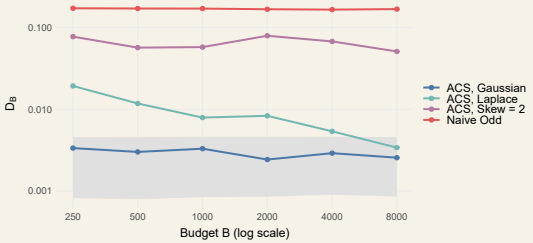


Notes: Design-conditional coverage of $\hat{\mu}_{K_B^\pi, N_B^\pi} \pm 1.96 \hat{b}_{K_B^\pi, N_B^\pi} / \sqrt{K_B^\pi}$. Grey lines are Clopper–Pearson 95% intervals for each realized K_B^π. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 5{,}000$, $F = 20$, $V = 2$.

Simulations – Sharp Boundary



(a) $E[Z_B^\pi \mid C_j]$



(b) $D_B = \sup_{t \in \mathcal{T}} \max_j \left| \mathbb{E}[e^{itZ_B^\pi} \mathbb{1}_{C_j}] - \phi(t)\mathbb{P}(C_j) \right|$

Notes: $(C_j)_j$ are quintiles of the realized K^π. Grey band represents worst-case 95% range of D_B over B under exact conditional normality. Symmetric errors are $N(0,1)$, asymmetric errors are centered $\Gamma(1,1)$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2{,}000$, $F = 20$, $V = 2$.

Outline

Introduction and Motivation

Global and Static Policies

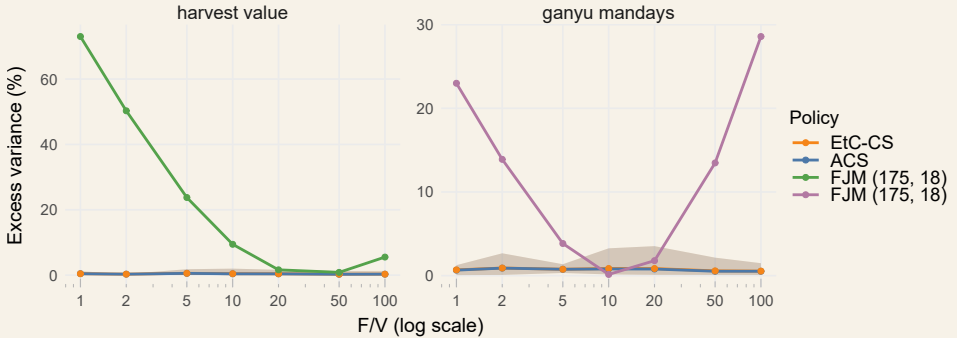
Adaptive Policies

Conditional Liberty

Application

Conclusion

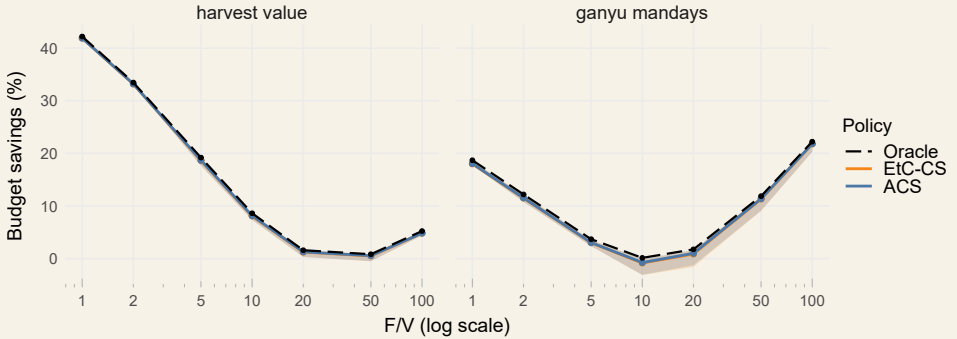
Fink et al (2020, AER)



Notes: Excess variance is presented relative to the oracle design (%). Lines are means over 10,000 simulations, bands are 95% intervals across simulations.

- Up to 41% (22%) **lower variance** at equal cost

Counterfactual Savings



Notes: Budget savings (%) relative to the imputed budget $B = 175\times(F+17.9V)$. 10^4 replications. 95% confidence intervals across simulations at selected budget.

- Up to 42% (22%) **lower cost** at equal variance

Inference and Standard Errors

	Paper	FJM	FJM	F/V			
	(Real)	(Real)	(Synthetic)	1	5	20	100
K/N	175/17.9	175/17.9	175/18	774/3	332/7	194/14	157/32
Cash	0.105*	0.112**	0.112***	0.109***	0.114***	0.112***	0.111***
	(0.041)	**(0.046)**	**(0.042)**	**(0.035)**	(0.039)	(0.042)	(0.040)
Maize	0.056	0.037	0.036	0.034	0.037	0.037	0.038
	(0.054)	**(0.047)**	**(0.042)**	**(0.035)**	(0.039)	(0.042)	(0.040)

Treatment-effect estimates for log(harvest value + 1), year-one outcomes.

Notes: Column 1 is reproduced from Fink et al. (2020), Table 4 Panel B (years one and two pooled). Counterfactual designs are generated by `ACS`; coefficients and standard errors are means over 1,000 simulations, and K/N reports the mean realized design. CR1 cluster standard errors, as in the original. Stars from the normal approximation: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Outline

Introduction and Motivation

Global and Static Policies

Adaptive Policies

Conditional Liberation

Application

Conclusion

Summary

- **Framework** for cost-optimal clustered designs under **unknown ICC**
- Two **adaptive algorithms**
  - negligible excess variance
  - dominate any misspecified static design, rate unimprovable
- **Adaptivity does not compromise inference** under **error symmetry**
  - asymptotically even designs do not disturb inference
  - no corrections, no discarded data
  - general principle beyond clusters as in neyman allocations
- Economically **meaningful**
  - 41% lower variance at equal cost, 42% lower cost at equal variance
  - runs live in the field (online survey coming up)

Thank you!

carlos.gonzalezperez@economics.ox.ac.uk

Carlos Gonzalez-Perez — and I am on the market!